RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

FDN 1204/ FDN 1306 Basic Mathematics
Tutorial 2

1. The sum of a number and its square is 72. Find the number.
2. If you square 5 more than a positive number, the result is 88 more than 12 times the number. Find the number
3. The sum of two numbers is 12 and the sum of their squares is 80. Find the numbers.
4. The length of a rectangle is 4 cm more than its width. The area of the rectangle is $96cm^2$. Find its dimensions.
5. The perimeter of a rectangle is 30 cm. Its area is $54cm^2$. Find the dimensions of this rectangle.
6. One number is 3 more than another. The sum of their squares is 149. Find the numbers.
7. The product of two consecutive positive numbers is 132. Find the numbers.
8. The sum of the squares of two consecutive positive even integers is 340. Find the integers
9. The length of a rectangle is 1 m longer than 3 times its width. The area is $80m^2$. Find the dimensions of this rectangle.
10. The length of a rectangle is 3 times its width. If the width is increased by 2 cm and the length is increased by 4 cm, its area will be doubled. Find the original dimensions of the rectangle.
11. The sum of a certain positive number and three times its square is 52. Find the number.
12. If one side of a square is increased by 10 cm and another side is increased by 5 cm, a rectangle is formed with an area 3 times the area of the square. Find the length of the side of the square.
13. If 12 is subtracted from 3 times the square of a positive number, the result is 5 times the number. Find the number.
14. Given two consecutive integers, if 3 times the first is increased by the square of the second, the result is 85. Find these numbers.
15. The perimeter of a rectangle is 60 m, its area is $200m^2$. What are the dimensions of the rectangle?
16. A negative number and its square add up to 6. Find the number.
17. We throw an object upward from the top of a 1200 ft tall building. The height of the object, (measured in feet) t seconds after we threw it is $h(t) = 16t^2 + 160\text{t} + 1200$.
    a) Where is the object 3 seconds after we threw it?
    b) How long does it take for the object to hit the ground
18. We are standing on the top of a 1680 ft tall building and throw a small object upwards. At every second, we measure the distance of the object from the ground. Exactly t seconds after we threw the object, its height, (measured in feet) is $h(t) = -16t^2 + 256\text{t} + 1680$.
    a) Find h (3). (h (3) represents the objectís position 3 seconds after we threw it.)
    b) How much does the object travel during the two seconds between 5 seconds and 7 seconds?

c) How long does it take for the object to reach a height of 2640 ft?

d) How long does it take for the object to hit the ground?

19. The Yellow Taxi Cab Co. charges a $2.75 flat rate and $.65 for each additional mile. Emma has no more than $14 to spend on a ride. How many miles can Emma ride without exceeding her spending limit?
20. Charlie has $500 in a savings account at the beginning of the summer, but withdraws $25 each week to spend on meals at Duchess (his favorite). He wants to have at least $200 in the account at the end of the summer for the start of school. How many weeks can Charlie withdraw money from the account?
21. The length of a rectangle is 4cm longer than the width. The perimeter is no more than 28cm. What are the maximum possible dimensions for the rectangle?
22. The sum of two consecutive integers is less than 55. Find the pair of integers with the greatest sum.
23. The sum of two consecutive even integers is at most 400. Find the pair of integers with the greatest sum. Practice: Use the 5 Step Process to solve each inequality word problem.
24. The sum of three consecutive odd integers is no less than 51. What is the middle integer?
25. Find two consecutive integers such that 7 times the smaller is less than 6 times the greater. What are the greatest such integers?
26. There are three exams given in a marking period. Ryan received an 85 and a 91 on the first two exams. What grade must he earn on the last exam in order to get an average of no less than a 90 for the marking period?
27. The length of a rectangle is 5 cm less than twice its width. If the perimeter of the rectangle is no more than 70cm, what are the maximum possible dimensions?
28. The length of a rectangle is 4cm more than three times its width. If the perimeter is more than 56 cm, what are the minimum possible dimensions of the rectangle (not allowing for fractional side measures)?